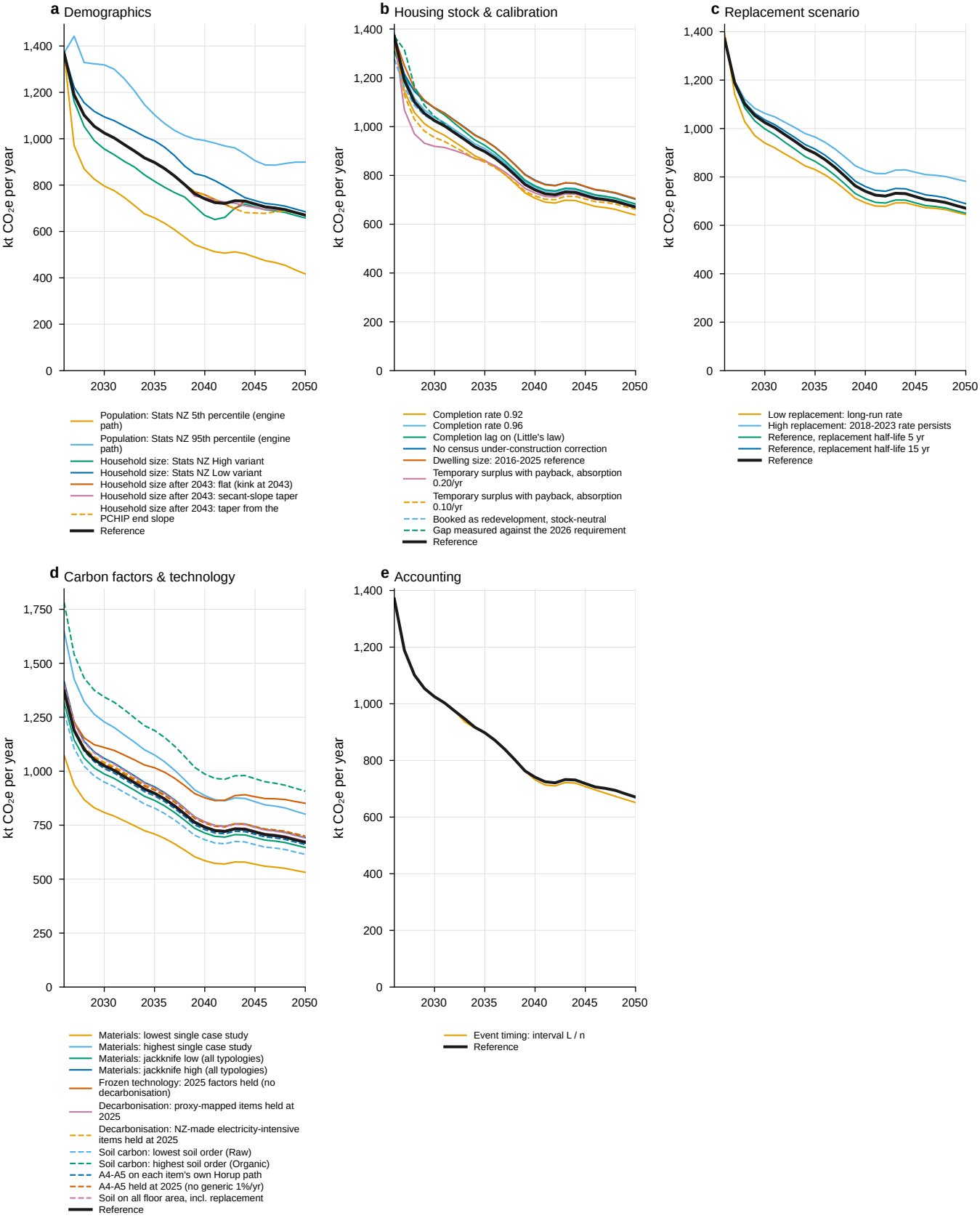
a Demographics
b Housing stock & calibration
c Replacement scenario
d Carbon factors & technology
e Accounting
kt CO2e per year
0
200
400
600
800
1,000
1,200
1,400
1,500
1,750
2030
2035
2040
2045
2050
Population: Stats NZ 5th percentile (engine path)
Population: Stats NZ 95th percentile (engine path)
Household size: Stats NZ High variant
Household size: Stats NZ Low variant
Household size after 2043: flat (kink at 2043)
Household size after 2043: secant-slope taper
Household size after 2043: taper from the PCHIP end slope
Reference
Completion rate 0.92
Completion rate 0.96
Completion lag on (Little's law)
No census under-construction correction
Dwelling size: 2016-2025 reference
Temporary surplus with payback, absorption 0.20/yr
Temporary surplus with payback, absorption 0.10/yr
Booked as redevelopment, stock-neutral
Gap measured against the 2026 requirement
Reference
Low replacement: long-run rate
High replacement: 2018-2023 rate persists
Reference, replacement half-life 5 yr
Reference, replacement half-life 15 yr
Reference
Materials: lowest single case study
Materials: highest single case study
Materials: jackknife low (all typologies)
Materials: jackknife high (all typologies)
Frozen technology: 2025 factors held (no decarbonisation)
Decarbonisation: proxy-mapped items held at 2025
Decarbonisation: NZ-made electricity-intensive items held at 2025
Soil carbon: lowest soil order (Raw)
Soil carbon: highest soil order (Organic)
A4-A5 on each item's own Horup path
A4-A5 held at 2025 (no generic 1%/yr)
Soil on all floor area, incl. replacement
Reference
Event timing: interval L / n
Reference